

*GATHERING DATA. TYPES OF STATISTICAL
STUDIES*

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INTRODUCTION

Statistical methods help us investigate questions in an objective manner. Statistical problem solving is an investigative process that involves four components:

- Formulate a statistical question
- Collect data
- Analyze data
- Interpret results

INTRODUCTION

The three main components of statistics for answering statistical questions are:

- **Design** refers to planning how to obtain relevant data that will efficiently shed light on the problem of interest. The design often involves taking a sample of the population of interest.
- **Description** means exploring and summarizing patterns in data.
- **Inference** means making decisions or predictions based on the data. It helps in deciding whether observed patterns are meaningful. Results of a study are considered **statistically significant** if they would rarely be observed with only ordinary random variation.

EXAMPLE: CELL PHONES AND YOUR HEALTH

How safe are cell phones to use? Cell phones emit electromagnetic radiation. There has been a growing concern over the potential health risks. Several studies have explored the possibilities of such risks:

- **Study 1:** A German study (Stang et al., 2001) compared 118 patients with a form of eye cancer to 475 healthy patients who did not have this cancer. The patient's cell phone use was measured using a questionnaire. On average, the eye cancer patients used cell phones more often.
- **Study 2:** A British study (Hepworth et al., 2006) compared 966 patients with brain cancer to 1716 patients without brain cancer. The second group was selected to match the first on age, gender and place of residence.
The patient's cell phone use was measured using a questionnaire. Cell phone use for the two groups was similar.

EXAMPLE: CELL PHONES AND YOUR HEALTH

- **Study 3:** A study by Volkow et al. 2011, published in *The Journal of the American Medical Association*, indicates that cell phone use speeds up activity in the brain. As part of a **randomized crossover study**, 47 participants were fitted with a cell phone device on each ear and then underwent two positron emission topography (PET) scans to measure a specific type of brain activity. During one scan, the cell phones were both turned off. During the other scan, an automated 50-minute muted call was made to the phone on the right ear. **Comparison** of the PET scans showed a significant increase in activity in the part of the brain closest to the antenna during the transmission of the automated call.

EXAMPLE: CELL PHONES AND YOUR HEALTH

Studies 1 and 3 found potential physiological responses to cell phone use. Study 2 did not.

- Things to notice
 - Different designs for gathering data
 - In every design we end up comparing *something* to *something else*.
 - Response variable: Studies 1 and 2: whether or not the subject has cancer. Study 3: The activity in the brain.
 - Explanatory variable: Studies 1 and 2: the amount of cell phone use. Study 3: Whether the cell phone is on or off.
 - Different types of data collected.
- Questions to explore
 - Why do results of different medical studies disagree?
 - What is the best study design for gathering data to explore whether cell phone use is associated with types of physiological activities in our bodies?

IN THIS LESSON ...

- The method of **comparison**
- Different designs for gathering data: observational studies and experiments

THE METHOD OF COMPARISON

Does treatment have an effect?

An ubiquitous problem in science, whether a treatment has an effect:

Treatment

- A magnetic field
- A metallic coating
- Welfare
- Decreasing the marginal tax rate
- A drug
- A fertilizer
- An advertising campaign

Effect

- To bend light
- To increase the durability of a part
- To decrease the crime rate
- To increase savings
- To relieve headaches
- To increase crop yields
- To increase sells

THE METHOD OF COMPARISON

To evaluate whether a treatment has an effect it is crucial to compare the outcome when treatment is applied (the outcome for the **treatment group**) with the outcome when treatment is withheld (the outcome for the **control group**), in situations that are as alike as possible but for the treatment. **This is called the method of comparison.**

How individuals come to be in the treatment group is important, too: This distinguishes experiments (or **experimental studies**) from **observational studies**, but we will see this later in more detail.

THE METHOD OF COMPARISON

- The method of comparison is arguably the most important idea in science: To determine whether a treatment has an effect, compare what happens with and without the treatment.
- It is a general and flexible tool for evaluating whether a treatment is effective, be it a drug, fertilizer, teaching technique, surgical procedure, quality control measure, car wax, or advertising campaign.
- The basic idea is to compare what happens with and without the treatment, to isolate the effect of the treatment. Only some of the individuals are treated, and the outcome for them is compared with the outcome for individuals who are not treated.

THE METHOD OF COMPARISON

- The group that receives treatment is called the treatment group, and the group that does not is called the control group.
- Sometimes, as is study 3, a single group's responses with (cell phone on) and without treatment (cell phone off) are compared (the individuals are their own controls).
- Sometimes, there are two or more treatment groups, and the responses of different groups are compared.
- Without controls, we have no idea whether the treatment is the cause of any observed response. With controls, it can be possible to determine whether treatment has an effect.

CONCEPTS REVIEW

- Observations and Variables. Types of variables.
- Population and samples.
- Sampling from a population. Anecdotal data.
- Explanatory and response variables.

STUDY DESIGNS FOR GATHERING DATA

Observational and Experimental studies (or experiments):

In Study 3 an experiment is performed. Subjects are assigned to experimental conditions (phone on or off) that we want to compare on the response outcome.

By contrast many studies merely observe the values on the response variable and the explanatory variable for the sampled subjects without doing anything to them. This is the case of studies 1 and 2.

STUDY DESIGNS FOR GATHERING DATA

There are two primary types of statistical studies: observational studies and experiments

- Observational studies: data is collected in a way that doesn't directly interfere with how data arise (e.g. surveys).
 - Can provide evidence on a naturally occurring association between variables, but they cannot by themselves show a causal connection.
- Experiments: Researchers randomly assign subjects to various treatments in order to establish causal connections between the explanatory and the response variable.

Observational Studies

OBSERVATIONAL STUDIES

- The researcher observes values of the response variable and explanatory variables for the sampled subjects, without anything being done to the subjects (such as imposing a treatment).
- The key difference between an experiment and an observational study is whether the researcher chooses who will receive the treatment.
- In an observational study, the researcher merely observes what happens to the treatment group, or compares those observations with observations of the control group, but the assignment to treatment or control is up to nature, historical accident, or the subjects themselves.
- *Choosing* is different from *classifying*: an investigator studying the effects of smoking on health can classify individuals as smokers or nonsmokers, but cannot choose who will smoke and who will not.

LURKING VARIABLES. CONFOUNDING.

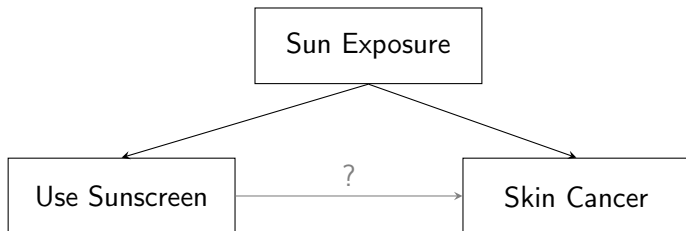
Observational studies are prone to confounding.

A **lurking variable** is a variable, usually unobserved, that influences the association between the variables of primary interest. In observational studies, lurking variables can affect the results. For instance, in Study 1, there could be lifestyle, genetic, or health differences between the subjects with eye cancer and those without it and between those who use cell phones a lot and those who don't, other than cell phone use. Computer use is a possible lifestyle lurking variable.

Lurking variables have the potential for **confounding**. When two explanatory variables are both associated with a response variable but are also associated with each other, confounding occurs. Confounding can hide a real effect, or can produce the spurious appearance of a treatment effect when the real cause is a difference between the treatment and control groups other than the treatment. Confounding can be a problem both in experimental and observational studies.

LURKING VARIABLES. CONFOUNDING.

An observational study tracks sunscreen use and skin cancer. It is found that the more sunscreen someone uses the more likely the person is to have skin cancer.



LURKING VARIABLES. CONFOUNDING.

Examples of lurking variables:

- Ice cream sales and drowning - lurking variable: temperature
- Reading level and shoe size - lurking variable: age
- Divorce rates and crime rates - lurking variable: time
- School testing for drugs and drugs use - lurking variable: school neighbourhood.

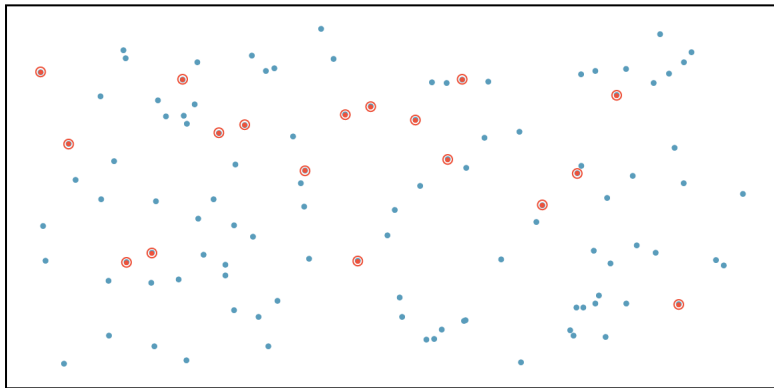
TYPES OF OBSERVATIONAL STUDIES.

- **Sample Surveys** select a sample of people from a population and collect data from them. They take a cross section of a population at the current time. Sample surveys are sometimes called cross-sectional studies or prevalence studies in Medicine.
- **Longitudinal studies** in which an inherent temporal dimension is present:
 - **Retrospective studies**, such as a case-control study, looks into the past; they collect data after events have taken place. Here, researchers review past events in medical records.
 - **Prospective studies** identify individuals and collect information as events unfold; they follow their subjects into the future.

SAMPLE SURVEYS. SAMPLING FRAME AND SAMPLING DESIGN

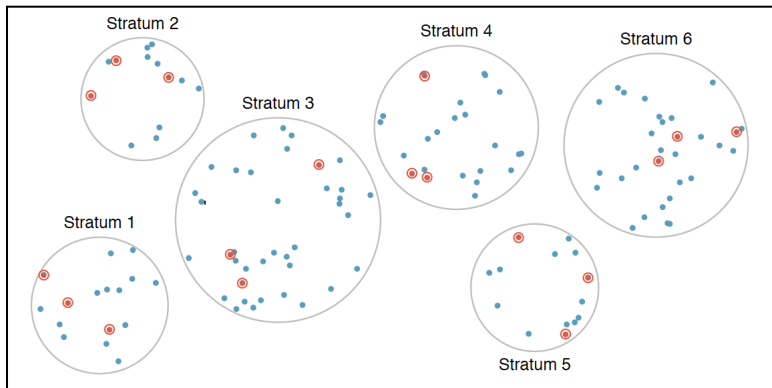
In sample surveys the **sampling frame** is the list of subjects in the population from which the sample is taken. Once you have a sampling frame, you need to specify a method for selecting subjects from it. The method used is called **sampling design** or **sampling method**. Let's consider here four random sampling techniques: **simple random sampling (srs)**, **stratified sampling**, **cluster sampling** and **multistage sampling**.

SAMPLE SURVEYS. SAMPLING FRAME AND SAMPLING DESIGN



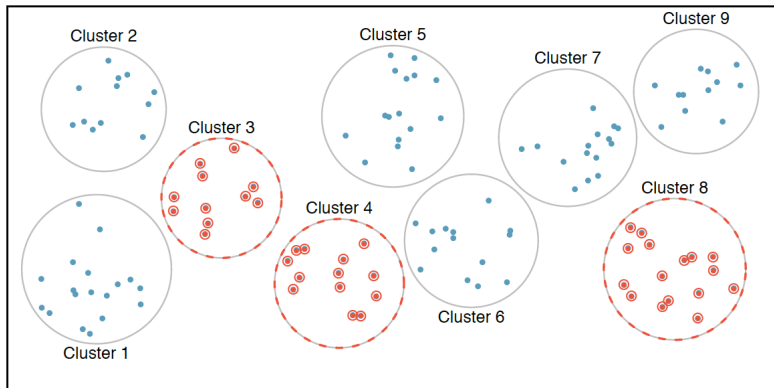
Simple random sampling used to randomly select 18 cases. Each case has the same probability of being selected.

SAMPLE SURVEYS. SAMPLING FRAME AND SAMPLING DESIGN



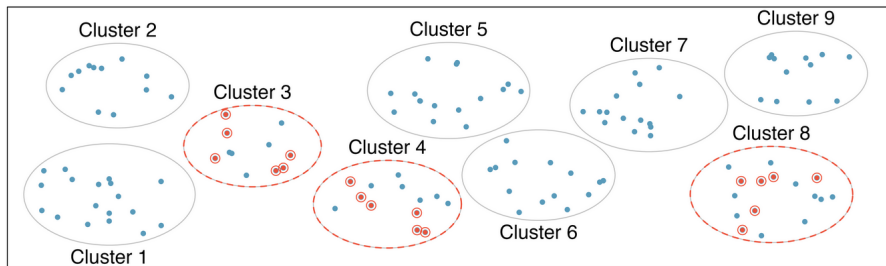
Stratified sampling: cases were grouped into strata made up of similar observations. Then, simple random sampling was employed within each stratum.

SAMPLE SURVEYS. SAMPLING FRAME AND SAMPLING DESIGN



Cluster sampling: data were binned into nine clusters. Clusters are usually **not** made up of homogeneous observations. We take a random sample of clusters and then sample all observations in that clusters.

SAMPLE SURVEYS. SAMPLING FRAME AND SAMPLING DESIGN



Multistage sampling: differs from cluster sampling only in that we randomly select a subset of each cluster to be included in the sample rather than measuring every case in each sampled cluster.

TYPES OF BIAS IN SAMPLE SURVEYS

When results from the sample are not representative of the population, they are said to exhibit **bias**. **Undercoverage** occurs when having a sampling frame that lacks representation from parts of the population.

- Sampling bias occurs from using nonrandom samples or having undercoverage.
- Nonresponse bias occurs when some sampled subjects cannot be reached or refuse to participate or fail to answer some questions. Even those who participate may not respond to some questions, resulting in nonresponse bias due to missing data.
- Response bias occurs when the subject gives an incorrect response (perhaps lying) or the way the interviewer asks the questions (or wording of a question in print) is confusing or misleading.

KEY PARTS OF A SAMPLE SURVEY

- Identify the population of all subjects of interest.
- Construct a sampling frame which attempts to list all subjects in the population.
- Use a random sampling design to select n subjects from the sampling frame.
- Be cautious about sampling bias due to nonrandom samples (such as volunteer samples) and sample undercoverage, response bias from subjects not giving their true response or from poorly worded questions, and nonresponse bias from refusal of subjects to participate.

POOR WAYS TO SAMPLE

Convenience samples are a type of survey sample that is easy to obtain relatively cheaply.

- They are unlikely to be representative of the population.
- They are prone to exhibit severe biases.
- Their results apply only to the observed subjects.

The most common type of **convenience sample** is **volunteer sample**.

- Subjects volunteer for the sample.
- Volunteers do not tend to be representative of the entire population.

A LARGE SAMPLE SIZE DOES NOT GUARANTEE AN UNBIASED SAMPLE

Many Internet surveys have thousands of respondents, but a volunteer sample of thousands is not as good as a random sample, even if that random sample is much smaller.

We're almost always better off with a simple random sample of 100 people than with a volunteer sample of thousands of people.

In fact, if the sample is random, the absolute size of the sample matters much more than the size relative to the population total.

When using a **simple random sample** of n subjects in estimating a proportion, the approximate margin of error is $\frac{1}{\sqrt{n}}$.

RETROSPECTIVE OBSERVATIONAL STUDIES

Rather than taking a cross section of a population at some time, such as with a sample survey, some studies are backward looking (**retrospective**) or forward looking (**prospective**). Observational studies in medicine are often retrospective.

A **case-control** study is a retrospective (observational) study in which subjects who have a response outcome of interest (the cases) and subjects who have the other response outcome (the controls) are compared on an explanatory variable. This is a popular design for medical studies in which it is not practical or ethical to perform an experiment.

CASE-CONTROL STUDIES: AN EXAMPLE

For instance, when studying lung cancer and smoking, the cases are subjects with lung cancer. The controls did not have cancer. The retrospective aspect refers to studying whether subjects had been smokers in the past. In this scenario, we can't randomly assign subjects into a smoking group and a nonsmoking group—this would involve asking some subjects to start smoking.

Smoker	Lung Cancer	
	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

STATISTICS THAT CAN DESCRIBE DATA IN 2X2 TABLES

An aside:

When you collect data on two binary categorical variables you form a 2x2 contingency table. You collect the counts, the number of cases presenting each combination of values for both variables. One of the variables can be regarded as being *explanatory* and the other as the *response* variable. To measure the strength of the association between both variables, there are several statistics that can be computed.

- Difference between proportions.
- Ratio of proportions: relative risk or risk ratio.
- Odds ratio.

	Belief in Afterlife?		
Gender	Yes	No	Total
Female	435	147	582
Male	375	134	509
Total	810	281	1091

STATISTICS THAT CAN DESCRIBE DATA IN 2X2 TABLES

An aside:

Difference between proportions might not be the best choice. Consider these two situations where success means you have gotten the illness, and success proportions are shown.

	Placebo	Treatment	Difference
Situation 1	0.5	0.45	0.05
Situation 2	0.1	0.05	0.05

Situation 1

	Experienced COVID?		
	Yes	No	Total
Placebo	50	50	100
Treatment	45	55	100

Treated saves 5 in 50, i.e., 1 in 10.

Situation 2

	Experienced COVID?		
	Yes	No	Total
Placebo	10	90	100
Treated	5	95	100

Treatment saves 5 in 10, i.e., 1 in 2.

STATISTICS THAT CAN DESCRIBE DATA IN 2X2 TABLES

An aside:

Consider this contingency table:

	Experienced COVID?		
Treatment	Yes	No	Total
Placebo (1)	112	8846	8958
J & J Vaccination (2)	32	8803	8835
Total	144	17,649	17,793

$$\hat{p}_1 = \frac{112}{8958} = 0.0125(1.25\%), \hat{p}_2 = \frac{32}{8835} = 0.0036(0.36\%)$$

$$\hat{p}_1 - \hat{p}_2 = 0.89\%$$

$$RR = \frac{\hat{p}_1}{\hat{p}_2} = 3.47$$

$$OR = \frac{odds_1}{odds_2} = \frac{\frac{\hat{p}_1}{1-\hat{p}_1}}{\frac{\hat{p}_2}{1-\hat{p}_2}} = 3.48$$

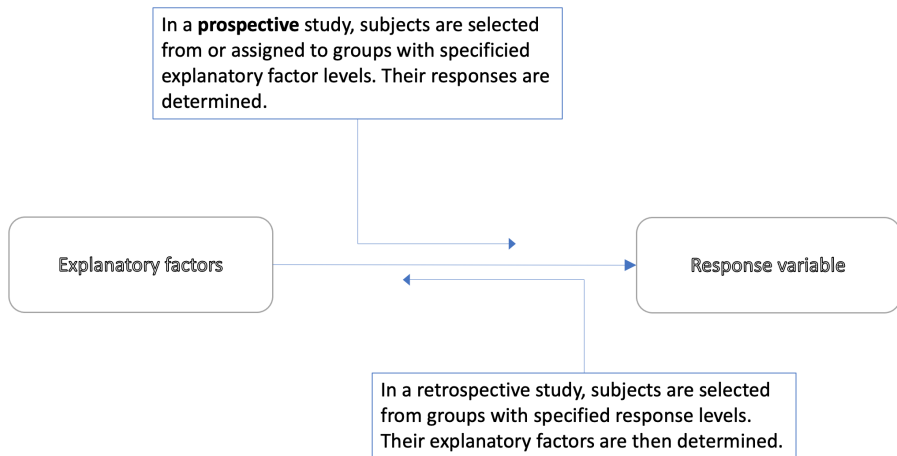
RETROSPECTIVE OBSERVATIONAL STUDIES

- In retrospective sampling, samples are taken for each level of the response variable and the levels of the explanatory factor are determined for subjects in these samples.
- An interesting property of the odds ratio is that it comes out the same regardless of whether you condition on the explanatory variable or on the response variable (e.g. it will be the same for prospective studies that condition on the exposure and for retrospective studies that condition on the outcome).
- The odds ratio is the only appropriate parameter (statistics) to estimate if the sampling is retrospective.

PROSPECTIVE OBSERVATIONAL STUDIES

A **prospective study** identifies a group (cohort) of people and observes them in the future. In research literature, prospective studies are often referred to as cohort studies.

RETROSPECTIVE VS. PROSPECTIVE OBSERVATIONAL STUDIES



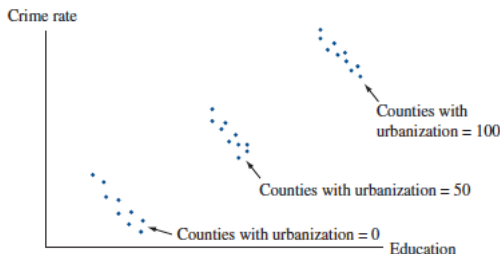
OBSERVATIONAL STUDIES AND CAUSATION

Key points with observational studies

- We can never definitively establish causation with an observational study.
- As more studies are done that adjust for confounding variables, the chance that a lurking variable remains that can explain the association is reduced.
- Unless a controlled experiment is performed, causal inferences are rarely warranted.
- Confounding is the rule, not the exception, unless the assignment to treatment and control is randomized.
- Beware of Simpson's Paradox: an observed association between two variables can *change or even reverse direction* when there is another variable that interacts strongly with both variables.

AN EXAMPLE OF SIMPSON'S PARADOX

The direction of an association between two variables can change after we include a third variable and analyze the data at separate levels of that variable.



Experiments

EXPERIMENTS

- An experiment deliberately imposes treatments on the experimental units (subjects: people, animals or other objects) in order to observe their responses.
- The goal of an experiment is to investigate the association – how the treatment affects the response.
- An advantage of an experimental study over an observational study is that it provides stronger evidence for causation.

1. Controlling

Researches assign treatments to cases and do their best to control any other differences in the group. A good experiment has a **control comparison group**. You can have also more than one categorical explanatory variable (**factors**). The combination of the different levels of each factor forms the treatment groups.

In human experiments, one way to reducing the bias is to conduct a **blind** experiment and a better way is to conduct a **double-blind** experiment.

- A placebo is a dummy treatment, i.e. sugar pill. Many subjects respond favorably to any treatment, even a placebo.
- A control group typically receives a placebo. A control group allows us to analyze the effectiveness of the primary treatment.

2. Randomization

Researchers randomize patients into treatment groups to account for variables that cannot be controlled. This tends to balance the comparison groups with respect to confounding variables.

3. Replication

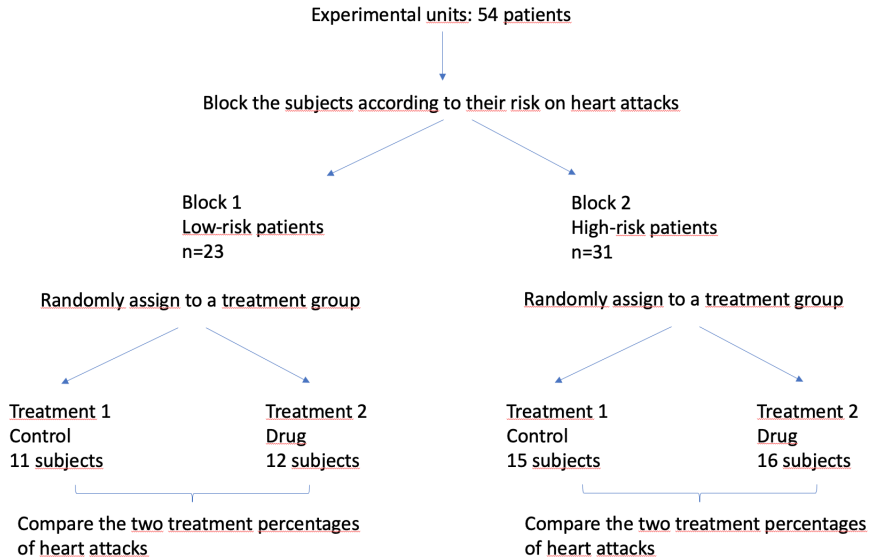
The more cases researchers observe, the more accurately they can estimate the effect of the explanatory variable on the response. In a single study, we replicate by collecting a sufficiently large sample. What is considered sufficiently large varies from experiment to experiment, but at a minimum we want to have multiple subjects (experimental units) per treatment group. Another way of achieving replication is replicating an entire study to verify an earlier finding and to increase confidence in the conclusions.

4. Blocking

Researchers sometimes know or suspect that variables, other than the treatment, influence the response. Under these circumstances, they may first group individuals based on this variable into blocks and then randomize cases within each block to the treatment groups. This strategy is often referred to as blocking. See an example on the next slide.

Block when you can; randomize when you cannot (George Box, 1919-2013).

BLOCKING



DIFFERENCE BETWEEN BLOCKING AND EXPLANATORY VARIABLES

- Blocking variables are characteristics that the experimental units come with, that we would like to control for.
- Blocking reduces the variability in the response variable and removes potential confounding caused by the blocking variable.
- Blocking is like stratifying, except used in experimental settings with random assignment, as opposed to sampling.

SUMMARY

Random sampling vs random assignment. Conclusions that can be drawn from data are dependent on how the data has been collected and how data has been allocated in control/treatment groups.

<i>ideal experiment</i>	Random assignment	No random assignment	<i>most observational studies</i>
Random sampling	Causal conclusion, generalized to the whole population.	No causal conclusion, correlation statement generalized to the whole population.	Generalizability
No random sampling	Causal conclusion, only for the sample.	No causal conclusion, correlation statement only for the sample.	No generalizability
<i>most experiments</i>	Causation	Correlation	<i>bad observational studies</i>

[Link to a similar figure with explanation, section 2.4.1 Summary](#)

EXERCISE 1

In an experiment studying a new drug, subjects were randomly assigned to either a placebo or an active drug. In addition, the method of delivery of the drug (pill, skin patch or nasal mist) was considered. How many factors and treatment groups were in this experiment?

EXERCISE 2

A study concluded that young adults who regularly played video games demonstrated better visual skills than young adults who don't play regularly. Sixteen young men volunteered to take a series of tests that measured their visual skills; those who had played video games in the previous six months performed better on tests than those who hadn't played.

- What are the explanatory and response variables?
- Was this an observational study or an experiment? Explain.
- Specify a potential lurking variable. Explain.

- Response variable: visual skills. Explanatory variable: whether or not the individual had played videogames before.
- It's an observational study.
- One possible lurking variable is reaction time. Excellent reaction times might make it easier, and therefore more fun, to play video games. Excellent reaction times also might lead young men to perform better on tasks measuring visual skills. These young men might have performed well on tasks measuring visual skills regardless of whether they played video games.

EXERCISE 3

Researchers collected data to examine the relationship between air pollutants and preterm births in Southern California. During the study air pollution levels were measured by air quality monitoring stations. Length of gestation data were collected on 143,196 births between the years 1989 and 1993, and air pollution exposure during gestation was calculated for each birth. (Ritz et al. 2000)

- Comment on whether or not the results of the study can be generalized to the population, and if the findings of the study can be used to establish causal relationships.

EXERCISE 4

A statistics student who is curious about the relationship between the amount of time students spend on social networking sites and their performance at school decides to conduct a survey. Various research strategies for collecting data are described below. In each, name the sampling method proposed and any bias you might expect.

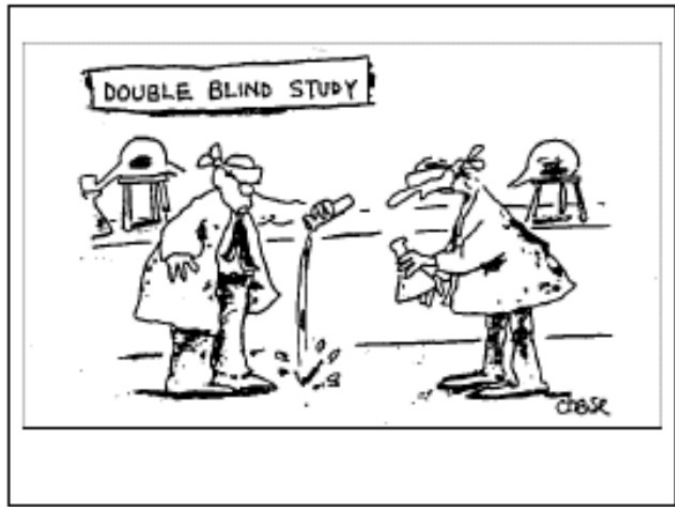
- They randomly sample 40 students from the study's population, give them the survey, ask them to fill it out and bring it back the next day.
- They give out the survey only to their friends, making sure each one of them fills out the survey.
- They post a link to an online survey on Facebook and ask their friends to fill out the survey.
- They randomly sample 5 classes and ask a random sample of students from those classes to fill out the survey.

EXERCISE 5

In a public health study on the effects of consumption of fruits and vegetables on psychological well-being in young adults, participants were randomly assigned to three groups: (1) diet-as-usual, (2) an ecological momentary intervention involving text message reminders to increase their fruits and vegetable consumption plus a voucher to purchase them, or (3) a fruit and vegetable intervention in which participants were given two additional daily servings of fresh fruits and vegetables to consume on top of their normal diet. Participants were asked to take a nightly survey on their smartphones. Participants were student volunteers at the University of Otago, New Zealand. At the end of the 14-day study, only participants in the third group showed improvements to their psychological well-being across the 14-days relative to the other groups. (Conner et al. 2017)

- What type of study is this? Identify the explanatory and response variables.
- Comment on whether the results of the study can be generalized to the population and on whether the results of the study can be used to establish causal relationships.

A BIT OF HUMOUR



- ▶ SticiGui. Chapter 28.
- The Art and Science of Learning from Data. Agresti and Franklin. 2013, Pearson Education, Inc.
- For the *Cell phones and your health* example:
 - Stang et al. (2001) The Possible Role of Radiofrequency Radiation in the Development of Uveal Melanoma. *Epidemiology*, 12(1), 7-12.
 - Hepworth et al. (2006) Mobile Phone Use and Risk of Glioma in Adults: Case-Control Study. *BMJ* 332.
 - Volkow et al. (2011) Effects of Cell Phone Radiofrequency Signal Exposure on Brain Glucose Metabolism *JAMA*, 305(8), 808-813.

- For the exercises:
 - Ritz, B. et al. (2000). "Effect of Air Pollution on Preterm Birth Among Children Born in Southern California Between 1989 and 1993." *Epidemiology* 11 (5): 502–11. [Link](#)
 - Conner, Tamlin S. et al. (2017). "Let them eat fruit! The effect of fruit and vegetable consumption on psychological well-being in young adults: A randomized controlled trial." *PloS One* 12 (2). [DOI](#)